

Introduction to Artificial Intelligence

Artificial intelligence is the simulation of human intelligence by machines. It was founded as an academic discipline in 1956 by John McCarthy. AI systems are used in many applications including speech recognition and image processing.

Machine Learning Basics

Machine learning is a subset of artificial intelligence. It allows computers to learn from data without being explicitly programmed. Common algorithms include linear regression, decision trees, and neural networks.

Deep Learning and Neural Networks

Deep learning uses neural networks with many layers to learn complex patterns. It has revolutionized computer vision and natural language processing. GPT and Claude are examples of large language models built with deep learning.

RAG Systems

Retrieval Augmented Generation combines document search with AI generation. RAG reduces hallucination by grounding answers in real documents. It is widely used in enterprise document Q&A systems.